



Review

Understanding the role of endocrine disrupting chemicals in testicular germ cell cancer: Insights into molecular mechanisms

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ABSTRACT

This comprehensive review examines the complex interplay between endocrine disrupting chemicals (EDCs) and the development of testicular germ cell tumors (TGCTs). Despite the high cure rates of TGCTs, challenges in diagnosis and treatment remain, necessitating a deeper understanding of the etiology of the disease. Here, we emphasize current knowledge on the role of EDCs as potential risk factors for TGCTs, focusing on pesticides and perfluorinated and polyfluoroalkyl substances (PFAs/PFCs). Evidence suggests that EDCs disrupt endocrine pathways and induce epigenetic changes that contribute to the development of TGCTs. However, the direct link between EDCs and TGCTs remains elusive and requires further investigation of the molecular mechanisms. We also highlighted the importance of studying nuclear receptors as potential targets for understanding TGCT etiology. In addition, recent evidence implicates PFAs/PFCs in TGCT incidence, highlighting the need for further research into their impact on human health. Overall, this review provides valuable insights into the potential role of EDCs in TGCT development and suggests avenues for future research, while also highlighting how understanding their influence may pave the way for novel therapeutic approaches to improve disease management.

1. Background

Testicular cancer is the most common cancer in young adult males (Siegel et al., 2023). Majority type of testicular cancer (95–98%) is testicular germ cell tumor (TGCT), which has several types comprising several histological subtypes (Albers et al., 2015; Berney et al., 2022). Initially identified in 1972 as carcinoma in situ of the testis (Skakkebaek, 1972), germ cell neoplasia in situ (GCNIS) has since been recognized as the precursor lesion leading to TGCTs (Berney et al., 2016). GCNIS-related TGCTs, classified as type II TGCTs, are the most common and malignant forms. They include seminomas arising from the germinal epithelium of the seminiferous tubules and non-seminomas. Non-seminomas may consist of pure or mixed elements, including embryonal carcinoma, teratoma, yolk sac tumor, and choriocarcinoma, with the possible presence of seminoma elements (Cheng et al., 2018). Most of TGCTs represents the abnormal development of physiological germ cells at different stages of maturation towards full spermatogenesis. GCNIS mimics embryonic germ cells as they migrate from the proximal region of the yolk sac to the genital ridge and function as primordial germ cells.

Similarly, upon reaching the genital ridge, germ cells, prior to their first maturation step, known as gonocytes, and prior to relocating to the spermatogonial niche, exhibit characteristics similar to embryonic germ cells. Consequently, during normal physiological development, these cells express a specific set of embryonic mRNAs, miRNAs and protein markers. These markers include the mast/stem cell growth factor receptor proto-oncogene c-KIT, placental alkaline phosphatase 1, octamer-binding transcription factor 4, nanog homeobox, the transcription factor SRY-box transcription factor 17, lin-28 homolog A, and the miRNAs *miR371-3* and *miR-302/367* (Looijenga et al., 2014; Rajpert-De Meyts et al., 2015).

In the western world, 0.01% of young adults suffer from testicular cancer (Bray et al., 2024). Despite its low prevalence in young adults, testicular cancer, predominantly TGCTs, poses significant challenges. Although TGCTs have a high overall cure rate of over 90% with current therapeutic regimens, current stratification tools often lead to both under- and overtreatment (Cheng et al., 2018). Certain patient subgroups have poor prognoses, and approximately 20–30% develop resistance to standard chemotherapeutic agents, resulting in a 50%

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survival rate and limited subsequent therapeutic options (Nakhaei-Rad et al., 2023). In addition, the diagnosis of TGCTs is a significant challenge. Traditional serum tumor markers such as alpha-fetoprotein (AFP), beta subunit of human chorionic gonadotropin (HCG), and lactate dehydrogenase (LDH), as well as imaging modalities, are not consistently reliable in the diagnosis and monitoring of TGCTs. These markers are elevated in only about 60% of TGCT cases and can also be elevated in several other medical conditions, leading to inaccuracies in diagnosis (Lobo and Leao, 2022). In addition, they are not very effective in detecting disease recurrence. Scrotal ultrasound, while confirming the presence of a testicular mass, is unable to accurately differentiate malignant TGCTs from other causes of testicular enlargement (Cheng et al., 2018). As a result, definitive diagnosis often relies on surgical removal and subsequent histopathologic examination.

Typically, global incidence of TGCTs has risen over the past two decades (Bergstrom et al., 1996; Ekblom and Akre, 1998; McGlynn et al., 2003; Trabert et al., 2015). However, the incidence of TGCTs varies between countries and ethnic groups. It appears to be higher in regions such as Scandinavia, Switzerland and Germany compared with Africa, Asia and Latin America, where reports indicate a significantly lower incidence. In addition, Caucasians, particularly those of European descent, have a higher prevalence compared to African Americans (Znaor et al., 2014). The exact reasons for these differences remain unclear. However, it is speculated that the increased incidence in Western countries may be due to greater exposure to etiological factors. Furthermore, the increase in incidence in different populations has been observed in cohorts born after 1920, a period that coincides with the introduction of several endocrine disrupting chemicals (EDCs) into commercial products. (Laws et al., 2021). In addition, TGCTs are recognized as hormone-sensitive diseases, further emphasizing the importance of understanding the role of EDCs in their development and pathology (Ferlin and Foresta, 2014).

Therefore, understanding the risk factors associated with the development of TGCT, particularly those related to EDC exposure, holds great promise for reducing the global incidence of these diseases (Skakkebaek et al., 2001, 2016). Furthermore, understanding the molecular aspects, including TGCT development induced by EDC exposure and the signaling pathways involving xenobiotic receptors, may provide novel diagnostic markers for specific TGCT types and potential therapeutic targets. This review aims to present the latest data on specific EDCs as risk factors for TGCTs and to explore the molecular mechanism of etiology and chemotherapy resistance of TGCTs induced by EDC exposure.

2. Risk factors that cause TGCTs

The pathogenesis of TGCTs is strongly influenced by environmental factors, although the specific exogenous risk factors remain poorly characterized. Several medical conditions such as cryptorchidism (Ferguson and AgoulNIK, 2013), hypospadias (Trabert et al., 2013), previous testicular cancer (Fossa et al., 2005), and impaired spermatogenesis (Moller and Skakkebaek, 1999) have been identified as potential risk factors for TGCTs. In addition, factors such as low maternal parity, twin birth, low birth order, breech presentation, and breastfeeding have also been studied for their association with TGCT risk (Cook et al., 2010; Richie, 2005; Sverdlow et al., 1997). While the etiologic factors related to medical conditions and TGCT pathogenesis have been extensively reviewed in previous articles (Cheng et al., 2018; Rajpert-De Meyts et al., 2016), the understanding of exogenous risk factors, including exposure to EDCs, remains limited. Here we highlight EDCs as a significant risk factor for TGCTs, particularly considering the increasing global incidence of the disease. Furthermore, by exploring recent evidence from case-control studies and examining the molecular aspects of EDCs in TGCT etiology, this review aims to improve researchers' understanding of TGCT pathogenesis and contribute to the improvement of TGCT treatment strategies.

3. Chemotherapy resistance of TGCTs

While TGCTs have an exceptionally high cure rate, unparalleled among solid tumors, their treatment presents a significant challenge due to chemoresistance. Approximately 10–15% of patients with primarily advanced TGCTs and 3–5% of all GCT patients fail platinum-based chemotherapy and ultimately succumb to their disease (Gandaglia et al., 2014; International Prognostic Factors Study et al., 2010). The development of cisplatin resistance in TGCTs has been identified as the underlying cause of such chemotherapy failure. Patients who experience progression during or within one month after completion of initial platinum-based chemotherapy, or who relapse/progress after second-line platinum-based treatment, are classified as "platinum-refractory" (Allen et al., 2017). The management of platinum-refractory TGCTs remains a major challenge for clinicians and researchers. In addition, the effects of xenoestrogens on chemoresistance in hormone-sensitive cancers (Wang and Tsai, 2012) may increase interest in exploring the effects of EDCs on chemoresistance in TGCTs.

4. EDC exposure and testicular cancer

EDCs are xenobiotic chemicals that interfere the endocrine homeostasis and cause adverse effect to human and their progeny (Encarnacao et al., 2019; Gore et al., 2015; Zoeller et al., 2012). Exposure to EDCs is inevitable because it is a substance of our environment, food, and products. EDCs play a pivotal role in disrupting the human endocrine system by modulating the synthesis, storage, release, transport, metabolism, binding, action, and/or excretion of endogenous hormones (Lobo and Leao, 2022). There is considerable evidence that EDCs pose a significant public health concern (Sifakis et al., 2017).

Both epidemiologic and clinical evidence strongly suggest that environmental and genetic factors contribute significantly to the initiation and progression of TGCTs by disrupting the normal differentiation processes of primordial germ cells. Early developmental stages, including embryonic, fetal, and infant stages, are particularly vulnerable to environmental exposures (Gilbert, 2005). Some studies suggest that exposure to EDCs during fetal development and in young males could induce genital malformations (Ferguson and AgoulNIK, 2013; Sifakis et al., 2017). While some epidemiologic studies have produced conflicting results, it is studied that factors such as dose, duration of exposure, developmental stage of the individual, maternal lifestyle, genetic predisposition, and variability in susceptibility to EDC exposure may collectively contribute to TGCTs (Ferguson and AgoulNIK, 2013; Sifakis et al., 2017; Soto and Sonnenschein, 2010).

4.1. Pesticides

Pesticides are formulated to target the nervous or reproductive systems of pests, often by disrupting their endocrine systems to eliminate them. These chemicals pose risks to humans because of the similarities between insect and human endocrine systems. Increased exposure to Dichlorodiphenyldichloroethylene (p,p'-DDE) has been associated with the risk of both seminoma and non-seminoma TGCTs (McGlynn et al., 2008). This case-control study analyzing prediagnostic serum samples from 754 cases and 928 controls enrolled in the US Servicemen's Testicular Tumor Environmental and Endocrine Determinants for Dichloro-Diphenyl-Trichloroethane exposure found that individuals in the highest serum p,p'-DDE quartile had an increased risk of TGCT compared with those in the lowest quartile 71% ($P = 0.002$), suggesting an association between DDE exposure and TGCT risk. In contrast, another case-control study of 876 adult men in Washington State, USA, found no association between serum DDE concentration and TGCT ($P = 0.36$) (Biggs et al., 2008).

Not only for the male patient himself, pesticide exposure to the pregnant woman had an effect on the incidence of TGCT in the son (Brauner et al., 2021). A 40-year follow-up study of more than 20,000

pregnancies in northern California showed that the ratio of maternal serum levels of 1,1,1-trichloro-2,2-bis(p-chlorophenyl)ethane to p,p'-DDE in the early postpartum period was associated with an increased risk of TGCTs (Odd ratio, OR: 3.56; 95% confidence interval, CI: 1.34–11.88; $P < 0.01$) (Cohn et al., 2010). A recent case-control study in California showed that exposure to certain pesticide ingredients during adolescence affects the risk of TGCT (Swartz et al., 2022). They found that nearby application of acephate in the year before a child's birth increased the risk of TGCT in adolescents regardless of the race of the study population (OR: 1.09; CI: 1.02–1.16). In addition, nationwide case-control studies in France reported associations between pesticide exposure and TGCT risk in offspring (Danjou et al., 2021; Paul et al., 2023). In the study by Danjou et al. domestic use of fungicides was associated with an increased risk of non-seminoma TGCT in the offspring (OR: 1.92; CI: 1.12–3.30). In addition, certain parental occupations with high pesticide exposure were positively associated with TGCT risk. A study of 9,569 cases and 32,028 controls from Denmark, Finland, Norway and Sweden found that the association between parental pesticide exposure before a child's birth and TGCT risk varied between countries, with an increased risk observed in Denmark (OR: 2.98; CI: 1.61–5.52) but not in other countries (Le Cornet et al., 2015).

4.2. Perfluorinated and polyfluoroalkyl substances (PFCs/PFAs)

PFCs/PFAS are a rapidly growing class of endocrine disrupting chemicals and widely used in a variety of human applications, including surfactants, non-stick coatings, waterproof fabrics, and components in firefighting foams (Di Nisio et al., 2019). These chemicals have attracted attention because of their ubiquitous presence in the environment, their long half-lives, and their potential adverse effects on human health (Nicole, 2020). Since the identification of the endocrine disrupting activity of PFCs/PFAs through in silico modeling in 2017 (Kar et al., 2017), there has been an increasing focus on assessing their public health implications (Coperchini et al., 2020). Numerous cohort studies using different experimental designs and conducted in different regions have attempted to investigate the possible association between exposure to PFCs/PFAs and testicular cancer (Gilliland and Mandel, 1993; Leonard et al., 2008; Li et al., 2022; Mastrantonio et al., 2018; Raleigh et al., 2014; Steenland and Woskie, 2012). The lack of association between testicular cancer and PFCs/PFAs found in these studies can be attributed to several limitations. First, the low mortality rate of testicular cancer results in a limited number of cases. In addition, the small number of cases exacerbates this problem, resulting in insufficient statistical power to detect potential associations between PFC/PFA exposure and testicular cancer risk. Another limitation is that many of these studies did not focus specifically on certain types of testicular cancer, such as TGCTs. Different types of testicular cancer may have different etiologies and risk factors, so failure to differentiate between them could mask potential associations between PFCs/PFAs exposure and specific subtypes of testicular cancer. As evidence, recent case-control study among U.S. Air Force Servicemen had reported the association between specific type of PFCs/PFAs level in serum and TGCTs. (Purdue et al., 2023). In this case, elevated serum levels of perfluorooctane sulfonic acid (PFOS) were associated with an increased risk of TGCTs (OR: 2.6; CI: 1.1–6.4; $P = 0.02$). Therefore, future research focusing on specific types of testicular cancer, particularly TGCTs, may provide valuable insights into the relationship between PFCs/PFAs exposure and testicular cancer risk.

As shown from human patient data, specific EDCs has association with TGCTs incidence and some of them are already recognized as carcinogenic in testis by WHO, European Commission and The Endocrine Society (Modica et al., 2023). The variation in results across data sets underscores the importance of considering factors such as dose, duration of exposure, region, and the specific type of EDC involved. This variability highlights the need for comprehensive research to delve deeper into the underlying causes of how EDCs induce TGCTs. In addition, because of the multi-exposure of EDCs in the environment, it is

difficult to reflect the real-life condition to study the mechanisms of how EDCs contribute to the biology of TGCTs. While the exact mechanisms by which EDCs contribute to testicular cancer are still being elucidated, evidence suggests that reducing exposure to these chemicals may help reduce the risk of developing TGCTs.

5. Molecular aspects of TGCTs related to EDCs exposure

EDCs can interact with multiple targets in the body, including hormone receptors, hormone synthesis and clearance processes, hormone transport pathways, signaling pathways, and epigenetic mechanisms (Macedo et al., 2023). Through these mechanisms of action, they exhibit diverse modes of action, such as activating or inactivating hormone receptors, stimulating or inhibiting hormone synthesis, disrupting signal transduction in hormone-responsive cells, and inducing epigenetic modifications in cells that produce or respond to hormones. Given that the risk of developing TGCT is closely linked to hormones (Douglas et al., 2006; Ferlin and Foresta, 2014), it is clear that EDCs have a significant link to its etiology. Because of the important role of sex steroid hormones in urogenital development, some researchers suggest that the in utero hormonal environment, which is highly sensitive to EDCs, may be related to the development of TGCTs (Rajpert-De Meyts, 2006; Skakkebaek et al., 2001). Morimoto et al. showed weak association with neonatal testosterone and risk of TGCTs in non-Hispanic whites ($P = 0.04$) (Morimoto et al., 2018). In this study, measurements of sex steroid hormones focused specifically on the window immediately after birth. During this period, the hormonal environment is a combination of those produced by the newborn testis and residual maternal hormones. It's worth noting that maternal exposure to EDCs can affect hormonal homeostasis during this critical period. Considering the importance of hormones in the etiology of TGCTs, understanding the molecular mechanisms by which EDCs affect hormone regulation will drive research and development of innovative strategies for the care and treatment of TGCTs. In addition to disrupting hormone homeostasis, EDCs can also induce epigenetic changes such as the methylation of cytosine residues on DNA, post-translational modification of histones, and non-coding RNA expression that may increase the risk of developing TGCTs.

5.1. Receptor mediated mechanism

The European Association of Urology define AFP, HCG, and LDH as routine serum markers of TGCT (Albers et al., 2015), which indicates the potential involvement of hormonal pathways in tumor development. In addition, researchers in this field have sought to explore the intricate mechanisms underlying hormonal disruption and the development of TGCTs, suggesting a complex interplay between hormones and tumorigenesis (Grenache, 2020).

The pathogenesis of TGCT is thought to be initiated during the fetal development of the testis (Ekbom, 1998). Hormonal imbalances, often influenced by EDCs, can lead to misleading signals altering cell phase-switching processes, potentially contributing to neoplastic transformations in adult life (Rajpert-de Meyts and Hoei-Hansen, 2007). EDCs are known acting through their interaction with nuclear receptors such as the estrogen receptors α and β (ER α and ER β), the androgen receptor (AR), the pregnant X receptor (PXR) and the peroxisome proliferator activated receptors α and γ (PPAR α , PPAR γ) (Balaguer et al., 2017; Grimaldi et al., 2015; le Maire et al., 2010). More recently, the retinoid X receptors (RXR α , RXR β and RXR γ) and the constitutive androstane receptor (CAR) have also been identified as targets of EDCs (Toporova and Balaguer, 2020).

In the realm of fetal development, the interplay between estrogen and androgen signaling pathways serves as a fundamental orchestrator of testicular development. The two predominant isoforms of the ER, ER α and ER β , respond to 17 β -estradiol during physiological signaling and intricately regulate cellular processes. However, the estrogenic potency

of several pesticides (Echeverri-Jaramillo et al., 2021; Gea et al., 2022; Legler et al., 1999, 2002) and perfluorooctanoic acid (PFOA), PFOS, perfluorononanoic acid, and perfluorodecanoic acid (Benninghoff et al., 2011), has been well documented, highlighting the vulnerability of estrogen receptors to environmental disruptors. Conversely, the influence of EDCs extends to ARs, with studies demonstrating both activation and antagonistic effects on AR signaling pathways. For example, organochlorine pesticides have been shown to activate the androgen receptor (Lemaire et al., 2004), while PFOA and PFOS act as antagonists in human cells (Di Nisio et al., 2019). Activation of the PXR, PPAR α , and CAR by certain pesticides (Abass et al., 2012; Kojima et al., 2011) and PFAs/PFCs (Bjork et al., 2011; Kublbeck et al., 2020) has been documented in both mouse and human studies. For example, pesticides have been reported to activate mouse and human PXR, while PFOA can mediate activation of PPAR α and PXR, as well as CAR. These receptors play a critical role in regulating the expression of cytochrome P450 (CYP) enzymes, including CYP2B, CYP2C, CYP3A (targets of both PXR and CAR) (Maglich et al., 2002), and CYP4A (primarily induced by PPAR α) (Kroetz et al., 1998). In addition, they regulate phase II enzymes such as uridine 5'-diphospho-glucuronosyltransferases (UGTs), glutathione S-transferases, and sulfotransferases (Kodama and Negishi, 2013; Schraplau et al., 2015). Activation or inactivation of these receptors can indirectly disrupt endocrine function by promoting the metabolism and inactivation of hormones. For example, CAR agonists have been shown to decrease thyroid hormone levels by inducing UGTs, while PPAR α agonists can decrease serum estradiol levels by inhibiting aromatase gene expression, thereby preventing the conversion of testosterone to estrogen (Lovekamp-Swan and Davis, 2003).

Estrogen acts delicately on the developmental stage, guiding cell fate and differentiation. But when its rhythm is disrupted, either by endogenous imbalances or exogenous perturbations, the subtle development of fetal testis can falter. Similarly, androgens exert their influence with measured force, sculpting the nascent testicular landscape with precision. However, should their homeostasis be disrupted, whether by aberrant synthesis or external perturbation, the intricate process of fetal testis development may stumble, potentially paving the way for TGCT development. Indeed, the delicate interplay of hormones, any discordance may serve as a initiation to the emergence of TGCTs, underscoring the profound impact of hormonal perturbations on fetal germ cell development and the imperative to unravel their intricacies.

The expression of receptors for EDCs in the pre- and neonatal testis underscores their potential role in developmental processes (Rouiller-Fabre et al., 2015). In addition, studies have linked exposure to EDCs to changes in sex hormone concentrations during pregnancy (Derakhshan et al., 2023). While the precise mechanism underlying the development of TGCTs due to EDC exposure remains elusive, it is clear that EDCs exert adverse effects on fetal development through receptor-mediated mechanisms. These effects may occur through both direct effects on fetal receptors and indirect effects through maternal receptors. Such perturbations may contribute to the formation of GCNIS, which could potentially progress to TGCTs, although the specific factors driving this progression remain unidentified (Dieckmann and Pichlmeier, 2004).

5.2. Epigenetic alteration

Epigenetic regulatory mechanisms play a critical role in both the initiation and maintenance of pluripotency in embryonic stem cells and in the maintenance of the identity of differentiated cell types (Boland et al., 2014). Disruption of these processes can lead to changes in chromosomal stability, stem cell properties, self-renewal capacity, and differentiation potential, which may ultimately contribute to the initiation and progression of cancer, including testicular cancer. Therefore, dysregulated or impaired epigenetic regulation may represent one of the underlying factors influencing the origin and biology of TGCTs. Alterations in DNA methylation patterns, particularly with respect to tumor

suppressor genes, are commonly observed in various cancers (Ellis et al., 2009) and are considered to be critical steps in tumorigenesis. Abnormal hypermethylation of CpG islands has been reported in numerous tumors, including TGCTs (Lind et al., 2007).

Studies in mouse has uncovered transgenerational effects of phthalates, showing that prenatal exposure can lead to epigenetic changes across multiple generations (Wen et al., 2020; Wu et al., 2010). Specifically, genes such as *Dnmt1*, *Tet1*, and *Tet2* have been found to have altered expression patterns. In addition, biopsies of CIS and various types of TGCTs from patients have demonstrated expression of DNMT1, TET1 and TET2 in all samples (Kristensen et al., 2014). It is interesting to note that TGCTs uncommonly exhibit aberrant methylation of tumor-related genes, unlike many other types of cancer (Okamoto and Kawakami, 2007). The expression pattern of DNMT3b has been deeply studied in stage I testicular seminoma (Arai et al., 2012). Embryonal carcinomas also showed high-level expression of DNMT3b (Almstrup et al., 2005; Beyrouthy et al., 2009; Biermann et al., 2007). Methoxychlor, a banned pesticide, is known to increase the expression of *Dnmt3b* in female reproductive organs (Zama and Uzumcu, 2009). This effect may also occur in the testis, possibly leading to the development of TGCTs.

Aberrant DNA methylation and overexpressed DNMTs were observed in TGCTs or its different subtypes (Okamoto and Kawakami, 2007). Surprisingly, each subtype of TGCTs has a distinct methylation pattern. Seminomas are characterized by a globally hypomethylated genome, whereas the more differentiated non-seminomas (such as teratoma, yolk sac tumor, and choriocarcinoma) exhibit a hypermethylated genome similar to somatic tissues (Koul et al., 2002; Smiraglia et al., 2002). Interestingly, embryonal carcinoma cases show an intermediate methylation pattern (Wermann et al., 2010). However, a small subset of seminomas, GCNIS and gonadoblastomas show a high level of DNA methylation similar to that observed in non-seminomatous tumors (Netto et al., 2008). EDCs of different types and doses are known to have different effects on methylation (Alavian-Ghavanini and Ruegg, 2018). Therefore, extensive research is warranted to understand the etiology of TGCTs by further exploring this phenomenon.

In TGCTs, DNA hypomethylation of oncogenes can lead to their overexpression, which in turn may contribute to carcinogenesis. For evidence, a study has identified epigenetic modifications in the uterus of mice exposed in utero to bisphenol A, affecting gene expression and estrogen responsiveness. The hypomethylation of *Hoxa10* following early bisphenol A exposure altered developmental programming of gene expression (Bromer et al., 2010). It has been reported that *HOXA10* mutation in human TGCTs cell line is related to its proliferation (Chen et al., 2018). Loss of nuclear HOXA10 canonical function results in attenuated TP53 signaling and increased cell cycling through regulating the G2/M checkpoint. Epigenetic changes in genes involved in fetal tissue patterning such as *Hoxa10* can lead to cellular instability, which in turn can lead to carcinogenesis (Burgio et al., 2018).

In non-seminoma tumors, several promoter genes, including *BRCA1*, *MGMT*, and *HIC1*, are frequently hypermethylated (Koul et al., 2002). Interestingly, exposure to certain doses of EDCs has been implicated in inducing hypermethylation of *BRCA1* in breast cancer. For example, exposure to bisphenol A and tetrachlorodibenzo-p-dioxin has been associated with hypermethylation of the *BRCA1* gene (Papoutsis et al., 2010; Qin et al., 2012). In addition, zearalenone, a non-steroidal estrogenic pesticide, has been shown to disrupt hormonal balance. Research has shown a significant increase in global levels of 5-methylcytosine, an epigenetic marker, in human breast cancer cell lines exposed to zearalenone (Karaman and Ozden, 2019). In addition, changes in *HIC1* promoter methylation have been shown to influence micronuclei levels in individuals exposed to polycyclic aromatic hydrocarbons (Ruiz-Hernandez et al., 2015). These findings suggest that exposure to EDCs may induce promoter methylation changes in genes such as *BRCA1* and *HIC1*, which may alter the risk of TGCTs in a manner consistent with the influence of genetic variants of these genes on TGCTs risk.

Epigenetic modifications in core members of the pluripotency network, such as *NANOG* and Octamer-binding transcription factor 3/4 (Hoei-Hansen et al., 2005), as well as histone methyltransferases (Eckert et al., 2008), may also be involved in TGCT development, potentially influenced by exposure to EDCs. This underscores the importance of investigating how epigenetic dysregulation driven by EDCs contributes to TGCT pathogenesis, in addition to the distinct methylation patterns observed in different TGCT subtypes. While aberrant methylation of tumor-related genes is uncommon in TGCTs, hypomethylation of oncogenes due to EDC exposure may play a role in carcinogenesis, highlighting the need for comprehensive research into the etiology of TGCTs and the impact of EDCs.

Dysregulation of non-coding RNAs is known to be associated with the development of TGCTs. In particular, the miRNA clusters *miR-371-373* (Lobo et al., 2019; Voorhoeve et al., 2007) and *miR-302* (Das et al., 2019) are well-established markers of seminoma. However, direct associations between EDC exposure and miRNA expression have not been extensively studied, although in utero exposure to EDCs has been shown to affect miRNA expression (Tsamou et al., 2018). Among the comprehensive profile, specific miRNAs downstream of the tumor suppressor phosphatase and tensin homolog (PTEN) are dysregulated. Loss of *Pten*, a tumor suppressor gene, marks the transition from intratubular germ cell neoplasia to invasive TGCTs (Di Vizio et al., 2005), suggesting a link between prenatal EDC exposure and TGCTs. In addition, recent studies have suggested a role for long non-coding RNAs in TGCT etiology (Cao et al., 2023; Guo et al., 2022). Given that prenatal exposure to certain EDCs alter placental long non-coding RNA expression (Machtinger et al., 2018), further elucidation of this relationship is warranted.

5.3. Chemoresistance

The mouse double minute 2 (*MDM2*) gene serve as a primary negative regulator of the P53 protein and is critical for maintaining the stability of the P53 pathway (Oliner et al., 1992). Through its E3 ubiquitin ligase activity, MDM2 can bind to P53, leading to its ubiquitination, cytoplasmic translocation, and subsequent degradation by proteasomes. Consequently, MDM2 amplification, detected in various human malignancies including TGCTs, is associated with chemotherapeutic resistance (Haronikova et al., 2021). Overexpression of MDM2 contributes to this resistance through both P53-dependent and independent pathways, highlighting its importance in therapeutic response. In addition, mutations that inactivate the P53 tumor suppressor pathway, such as those leading to MDM2 amplification, have been implicated in promoting cisplatin resistance in TGCTs (Bagrodia et al., 2016). In addition, exposure to bisphenol A has been associated with P53 inactivation in human cancer cells, suggesting a potential role in chemoresistance mechanisms (Dairkee et al., 2013). These findings underscore the complex interplay between MDM2, P53 signaling and environmental exposures in modulating chemotherapeutic response, particularly in the context of TGCTs.

6. Perspectives

The global incidence of TGCTs is increasing despite their typically high cure rates. In this review, we highlight EDCs as potential risk factors contributing to the increasing incidence of TGCTs. Several studies have suggested associations between EDC exposure and TGCT development, highlighting the need to understand the potential molecular mechanisms involved. While the direct link between EDCs and TGCT development remains unclear, one possible mechanism involves disruption of endocrine pathways. In addition, EDCs may induce epigenetic changes that contribute to TGCT development by targeting genes known to be involved in TGCT etiology. However, understanding the link between EDCs and TGCTs is complicated by the developmental process of TGCTs, which arise from primordial germ cells during prenatal life. This makes it difficult to track hormonal changes during the

neonatal period in both human and animal studies, requiring the development of novel methodologies. Furthermore, studies investigating EDCs and TGCTs have yielded mixed results that are influenced by factors such as dose, duration of exposure, and developmental stage of exposure. Therefore, to elucidate the effects of EDCs on TGCTs, experimental designs should include specific types or mixtures of EDCs, relevant doses that reflect human exposure, and exposure timings that are critical for TGCT development. Emphasizing the link between EDCs and nuclear receptors may provide valuable insights into the unknown mechanisms underlying TGCT development. In addition, recent evidence suggests a potential link between PFAs/PFCs and TGCT incidence, with prenatal exposure to PFAs/PFCs adversely affecting offspring development and future fertility. This highlights the importance of studying PFA/PFCs and their receptors as potential targets for understanding the underlying mechanisms of TGCT etiology. Furthermore, elucidating the pathways by which EDCs exert their effects leading to TGCTs may also inspire novel approaches to disease treatment. This is of particular interest in the quest to improve cure rates, especially for individuals who are resistant to current treatment modalities.

CRedit authorship contribution statement

Won-Ki Pang: Writing – review & editing, Writing – original draft. **Ekaterina Kuznetsova:** Writing – review & editing. **Hélène Holota:** Writing – review & editing. **Angélique De Haze:** Writing – review & editing. **Claude Beaudoin:** Writing – review & editing. **David H. Volle:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare to have no conflict of interests.

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